

on board. There was also great progress in avionics and navigation systems. In the late 1980s, the Airbus A320 introduced digital fly-by-wire controls to civil aviation, as well as the so-called “glass cockpit.” The Boeing 777, introduced into service in 1995, was the first American fly-by-wire airliner.

Prospects for passenger travel

It was hard to feel that the chief executives of aircraft manufacturers or airlines in the 21st century had much in common with the flight-besotted empire builders of an earlier era. This tended to be a business like any other, dominated by marketing gurus and accountants. The kind of vision that produces entirely new aircraft ideas was in short supply. Although revisiting supersonic passenger flight or exploring even faster “hypersonic” designs was not ruled out, the accepted wisdom was that further leaps in aircraft size and range would make better economic sense. Airbus pressed ahead with a project for a super-jumbo in the face of considerable scepticism from air-industry experts. The A380 (see pages 430–31) defied its critics, overcoming considerable production problems to enter service in 2007 and then proving commercially viable.

Wrong-footed by the development and marketing of the Airbus A380, Boeing struggled to catch up with the launch of the 787 Dreamliner, which entered service after frustrating delays in 2011.

The key selling point of the 787 was neither speed nor size, but fuel efficiency. By 2016 projections were for future air passenger traffic doubling in volume to 7 billion journeys a year by 2034. There were constraints that might affect this expansion, particularly concerns about the

effect of aircraft on global warming. But this did not mean that any serious question mark hung over the future of mass air travel. A triumph of technology and organization, it had given human beings an undreamed-of mobility, and it was here to stay.



NOISE REDUCTION

The Boeing 787's engines have a chevron-toothed rim at the rear to reduce the noise caused by the jet exhaust meeting the outside air. Together with the use of sound-absorbing materials, this feature helps the 787 meet stringent requirements for quieter flight imposed on environmental grounds.

PASSENGER COMFORT

The 787 is a medium-size wide-body airliner, typically seating 234 to 296 passengers. Much effort has been put into improving the customers' experience of flight, including the installation of cabin windows that are larger than in previous airliners, providing a better view.



BOEING 787 DREAMLINER

Introduced into commercial service in 2011, the Boeing 787 Dreamliner is powered by two Rolls-Royce Trent 1000 or General Electric GEnx engines. Eighty per cent of its airframe consists of lightweight composite materials, helping make it an exceptionally fuel-efficient aircraft, with a maximum range of around 9,320 miles (15,000km). Sales were rapid by the standard of wide-body airliners, the 400th Dreamliner entering service in 2016.